

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Data

Define function for finding area

```
def sa(radiator_L, radiator_W, radiator_H, tube_L, tube_W, tube_H, tube_N, fin_L, fin_W, fin_H, fin_N):
    coolant_sa=2*(tube_L*tube_W+tube_L*tube_H)*tube_N
    air_sa=2*(fin_L*fin_W+fin_L*fin_H)*fin_N
    return coolant_sa, air_sa
```

Some info

```
m_dot_water=10/60 # kg/s
cp_water=4.186    # kJ/kgK
cp_air=1.006      # kJ/kgK
rho_air = 1.1614  # kg/m3

T_in_water=70     # degree C
T_out_water=60    # degree C
T_in_air=25       # degree C
```

```
r_l=0.2
r_w=0.02
r_h=0.095
t_l=r_l
t_w=r_w
f_l=r_w
f_w=0.002
f_h=0.008
f_n=990
t_n=14
t_h=(r_h-15*f_h)/t_n
```

```
coolant_sa, A_air=sa(r_l, r_w, r_h, t_l, t_w, t_h, t_n, f_l, f_w, f_h, f_n)
coolant_sa, A_air
```

```
(0.10200000000000001, 0.396)
```

Test case

1

```
v=4.3          # m/s
m_dot_air=v*rho_air*A_air # kg/s
q=m_dot_water*cp_water*(T_in_water-T_out_water)
T_out_air=T_in_air+q/(m_dot_air*cp_air)
```

q

6.976666666666667

T_out_air

28.506747771695807

T_out_air-T_in_air

3.5067477716958066

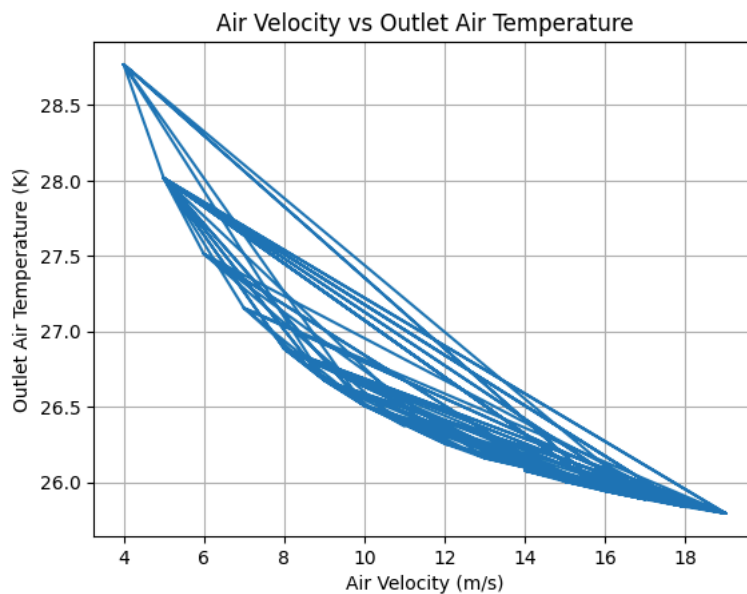
2

Double-click (or enter) to edit

```
v_arr=np.random.randint(4,20,100)
```

```
m_dot_air_arr=v_arr*rho_air*A_air
t_out=T_in_air+q/(m_dot_air_arr*cp_air)
```

```
plt.plot(v_arr, t_out)
plt.xlabel('Air Velocity (m/s)')
plt.ylabel('Outlet Air Temperature (K)')
plt.title('Air Velocity vs Outlet Air Temperature')
plt.grid()
plt.show()
```

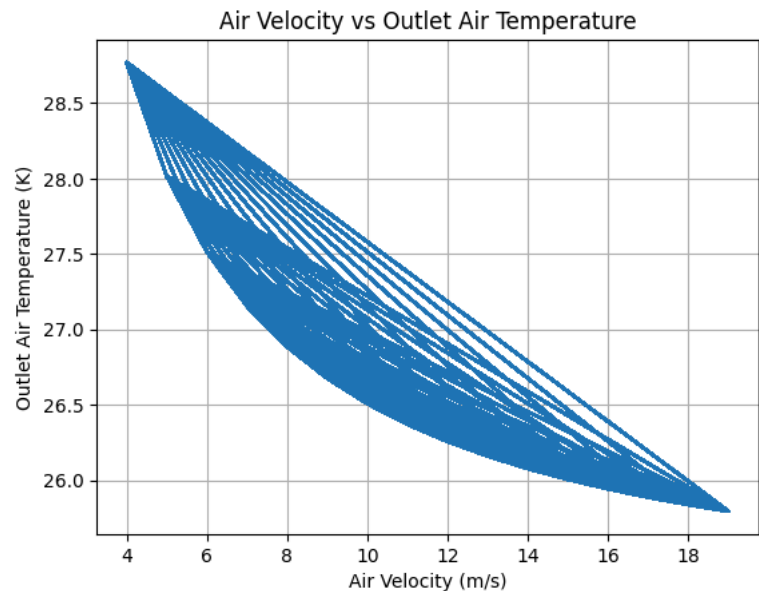


3

```
v_arr=np.random.randint(4,20,100000)
m_dot_air_arr=v_arr*rho_air*A_air
t_out=T_in_air+q/(m_dot_air_arr*cp_air)
t_out=np.array(t_out)
```

```
df=pd.DataFrame({v_arr})
```

```
plt.plot(v_arr, t_out)
plt.xlabel('Air Velocity (m/s)')
plt.ylabel('Outlet Air Temperature (K)')
plt.title('Air Velocity vs Outlet Air Temperature')
plt.grid()
plt.show()
```



Start coding or [generate](#) with AI.

black radiator

```
r_l=0.242
r_w=0.12
r_h=0.024
t_l=r_l
t_w=r_w
f_l=r_w
f_w=0.0015
f_h=0.009
f_n=95*2*13
t_n=12
t_h=0.0009
```

```
coolant_area,air_area=sa(r_l, r_w, r_h, t_l, t_w, t_h, t_n, f_l, f_w, f_h,f_n)
air_area=f_h*f_w*f_n
vel=10
T_in=40
m=vel*air_area*rho_air
Q=abs(vel*air_area*rho_air*cp_air*(8))
Q_w=m_dot_water*cp_water*(T_in_water-T_out_water)
```

m

0.38726883

air_area

0.033345

T=Q_w/(m)
T

18.015048271937264

Q

3.11673954384

Q vs ambient Temperature

```
T_air_out=50
```

```
T_amb=np.linspace(25,45,100)
Q_dot=rho_air*cp_air*(T_air_out-T_amb)*air_area
plt.plot(T_amb, Q_dot)
```

```
plt.xlabel('Ambient Temperature (K)')  
plt.ylabel('Q (W)')  
plt.title('Q vs Ambient Temperature')  
plt.grid()  
plt.show()
```

